

Relative Covariance

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This article describes how the phenotypic covariance between two relatives can be modeled. We consider different trait architectures, starting with the simplest and iteratively adding complexity. Assumptions are stated clearly where applicable.

1 AE w/ no AM

Consider a trait solely determined by an additive genetic component and an independent environment noise component:

$$y_i = g_i + e_i$$

Where we assume:

- y is standardized: $E[y] = 0$ and $\text{Var}[y] = 1$
- The genetic and environmental components are independent: $g \perp e$
- The genetic and environmental components are mean-centered: $E[g] = E[e] = 0$
- The variance of each component is given by: $\text{Var}[g] = V_A$ and $\text{Var}[e] = V_E$

Furthermore, **we assume the environment is completely random, even between relatives:** $e_i \perp e_j$ for all i, j . The expected covariance between two individuals i and j is given by:

$$\begin{aligned} E[\text{Cov}[y_i, y_j]] &= E[\text{Cov}[g_i + e_i, g_j + e_j]] \\ &= E[\text{Cov}[g_i, g_j]] \\ &= V_A E[\text{Corr}[g_i, g_j]] \end{aligned}$$

where the second line follows from assuming the environmental term is independent of genetics and across individuals.

1.1 Approximating genetic covariance by realized relatedness

We will now assume that the genetic component of a trait is the linear combination of individual variant allele counts, which are standardized:

$$g_i = \sum_k \beta_k x_{ik}$$

where β_k is the per-standardized-allele causal effect of variant k and x_{ik} is the standardized allele count of individual i for variant k . We can therefore expand the expected covariance of two individuals as:

$$\begin{aligned} E[\text{Cov}[y_i, y_j]] &= V_A E[\text{Corr}[g_i, g_j]] \\ &= V_A E[\text{Corr}[\sum_k \beta_k x_{ik}, \sum_{k'} \beta_{k'} x_{jk'}]] \end{aligned}$$

We assume no inbreeding and random mating, meaning that maternally- and paternally-inherited alleles are uncorrelated: $E[\text{Corr}[x_{ik}^{(m)}, x_{ik}^{(p)}]] = 0$.

We assume that all variants are physically unlinked and uncorrelated on a population level and that causal effects are independently distributed:

$$E[\text{Corr}[x_{ik}, x_{ik'}]] = 0 \text{ and } E[\text{Corr}[\beta_k, \beta_{k'}]] = 0 \text{ when } k \neq k'$$

This allows us to express the correlation in genetic values as the sum of correlations of individual variant genotypes:

$$E[\text{Corr}[g_i, g_j]] = \sum_k \beta_k E[\text{Corr}[x_{ik}, x_{jk}]]$$

For a particular variant, the expected correlation of two individuals' genotypes is given by their identical-by-descent (IBD) status.

Connect correlation between genotypes to their IBS state. Then express expected IBS state (and therefore correlation) as function of IBD, under no-mutation assumption below.

We assume that when two alleles are IBD, then they are also identical-by-state (IBS), i.e. no recent mutations. When two alleles are not IBD, we assume they are drawn independently from the population and therefore uncorrelated.

$$\begin{aligned} E[\text{Corr}[x_{ik}, x_{jk}] | \text{IBD}[x_{ik}, x_{jk}] = 0] &= 0 \\ E[\text{Corr}[x_{ik}, x_{jk}] | \text{IBD}[x_{ik}, x_{jk}] = 1] &= 0.5 \\ E[\text{Corr}[x_{ik}, x_{jk}] | \text{IBD}[x_{ik}, x_{jk}] = 2] &= 1 \end{aligned}$$

The genome-wide correlation in genetic value can therefore be expressed as:

$$E[\text{Corr}[g_i, g_j]] = \frac{1}{2} \Pr[\text{IBD}[x_i, x_j] = 1] + \Pr[\text{IBD}[x_i, x_j] = 2] = E[\pi_{ij}]$$

where π_{ij} is the realized relatedness between individuals i and j , i.e. the proportion of their genomes that is IBD. The relationship between genetic value correlation and realized relatedness should be expressed through expected values, since the true